

Zirui Li 李子睿

NTU AI-for-X Postdoc at Nanyang Technological University ([AutoMan Lab](#))
Incoming Marie Skłodowska-Curie (MSCA) Postdoctoral Fellowship at EPFL
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Brief Introduction

His research spans collective intelligence emergence in edge systems, data-driven interactive behaviour modelling, lifelong learning in robotics, and safety assessment, with extensions to healthcare and energy applications. (including co-first and first in student designation), He has published as first (or co-first) author 2 books, 17 peer-reviewed journal articles, and 3 conference papers. His work has been cited over 1,600 times. His other contributions include several patents, open-source software, and publicly available datasets, with a GitHub repository exceeding 1,000 stars.

Dr. Li has led and contributed to industrial projects with the National Natural Science Foundation of China, SAIC Motor, Baidu and Huawei, delivering applications in automated driving, behaviour analysis and public transport prediction. He received multiple academic awards, delivered 15 invited talks at international conferences, and served as (co-) chair or committee member for various international workshops and fora.

His international research experience includes visiting positions at Delft University of Technology (1 year) and Technische Universität Dresden (2 years), fostering collaborations with partners across US, Europe, Asia, and industry. He is also the working group chair for the IEEE P3439 standard on driving risk assessment and maintains an extensive global network of academic and industrial collaborators.

Web Identity

[Homepage](#) | [GoogleScholar](#) | [ResearchGate](#) | [LinkedIn](#) | [Github](#) | [Scopus ID 57204705157](#)

Education

Beijing Institute of Technology	Beijing, China	Sep. 2019–Jun. 2025
• Ph.D. in Mechanical Engineering, Intelligent Vehicle Research Center • Thesis: <i>Spatiotemporal Risk Continual Learning-based Interactive Motion Planning of AVs</i> • Supervisor: Prof. Jianwei Gong		
Technische Universität Dresden	Dresden, Germany	Jun. 2022–Jun. 2024
• Visiting Researcher, Chair of Traffic Process Automation • Supervisor: Prof. Meng Wang		
Delft University of Technology	Delft, The Netherlands	Jun. 2021–May 2022
• Visiting Researcher, Department of Transportation & Planning • Supervisor: Prof. Bart van Arem, Prof. Meng Wang and Dr. Victor Knoop		
Beijing Institute of Technology	Beijing, China	Sep. 2015–Jun. 2019
• Bachelor of Engineering, Xuteli Talent School (Mechanical Engineering) • Outstanding Graduate & Outstanding Undergraduate Thesis • Thesis: <i>Analysis of influencing factors for drivers' decision-making at urban intersections</i>		

Professional Experience

Nanyang Technological University, NTU AI-for-X Postdoc/AutoMan Lab	Mar. 2026-Now
• Project Mentor: Prof. Chen Lv from AutoMab Lab • AI Mentor: Prof. Ziwei Liu from MMLab • Goal: This project aims to realize lifelong spatiotemporal KS along the temporal dimension within each individual and along the spatial dimension across multiple embodied agents, ultimately enabling the emergence of collective intelligence.	

- Research grant: 250,000 SGD

Nanyang Technological University, Continental-NTU Postdoc Project/AutoMan Lab Sep. 2025–Feb. 2026

- Led development of situation awareness, interaction modelling and decision-making for automated vehicles.
- Tested rig set-up, and experimental validation.

Beijing Institute of Technology, NSFC/BIT-SAIC Project Sep. 2019–Jun. 2025

- Developed an interactive behaviour model, which can predict positions of traffic participants.
- Created a risk model for automated vehicle, which can continually learn and update.
- Developed a motion planner in dense environment with low computation cost and high safety.
- Developed a motion planner by integrating objective and passengers' subjective risk.
- Applied deployment and coordinated projects with NSFC/SAIC Motor/HUAWEI.

Delft University of Technology, CSC Project/SHAPE-IT Jun. 2021–Jun. 2022

- Conducted a [large-scale online survey](#) of perceived risk data (N=2,164).
- Developed two trajectory prediction model and rank 1st in [INTERPRET challenge](#).
- Developed a reachability analysis-based risk assessment model for automated model. ([News](#))

Technische Universität Dresden, CSC Project/VAMOS Jun. 2022–Jun. 2024

- Developed a high-performance model for bus/tram arrival time prediction.
- Constructed a dataset and benchmark for public transport arrival time prediction.
- Leading the IEEE standard development ([P3439](#)) as the working group chair.

Selected Publications (# Shared first authorship, * Corresponding author)

Books:

- [B1] Lu C, [Li Zirui](#), et al. Scene Understanding and Behaviour Prediction for Intelligent Vehicles. *Beijing Institute of Technology Press*, 2025. [In Chinese, National Key Publications in the 14th Five-Year Plan]
- [B2] Lu C, Gong J, [Li Zirui](#). Interactive Behaviour Prediction and Decision-making for Intelligent Vehicles. *Beijing Institute of Technology Press*, 2021. [In Chinese, National Key Publications in the 14th Five-Year Plan]
- Contribution:** The above two books presents our recent work in dynamic scene understanding, interactive behaviour prediction and intelligent decision-making technology systemically. And they are recognized as the “National Key Publications in the 14th Five-Year Plan”.

Key Journal Publications:

- [J1] [Li Zirui](#), et al. Interactive Behaviour Prediction for Heterogeneous Traffic Participants in the Urban Road: A Graph-Neural-Network-based Multitask Learning Framework. *IEEE/ASME Transactions on Mechatronics* 26.3 (2021): 1339-1349. [[Link](#)] [[73 citations](#)] [IF=7.3]
Contribution: We present the first multi-task interactive behaviour learning framework for Heterogeneous Traffic Participants in the Urban Road.
- [J2] [Li Zirui](#), et al. Importance Weighted Gaussian Process Regression for Transferable Driver Behaviour Learning in the Lane Change Scenario. *IEEE Transactions on Vehicular Technology*, 2020, 69(11): 12497-12509. [[Link](#)] [[42 citations](#)] [IF=7.1]
Contribution: A novel Gaussian process regression (GPR) importance weighted GPR is proposed for transferable driver behaviour modelling.
- [J3] [Li, Zirui](#), et al. A Hierarchical Framework for Interactive Behaviour Prediction of Heterogeneous Traffic Participants based on Graph Neural Network. *IEEE Transactions on Intelligent Transportation Systems*, 2021, 23(7): 9102-9114. [[Link](#)] [[81 citations](#)] [IF=8.4]
Contribution: This work propose a hierarchical GNN framework to model interactions of heterogeneous traffic participants combined with LSTM to predict their trajectories, which consists of two modules: interactive events recognition and trajectory prediction.
- [J4] [Li Zirui](#), et al. Personalized driver braking behaviour modelling in the car-following scenario: An importance-weight-based Transfer Learning Approach. *IEEE Transactions on Industrial Electronics*, 2022, 69(10): 10704-10714. [[Link](#)] [[31 citations](#)] [IF=7.2]
Contribution: An instance-level transfer learning framework is proposed to recognize braking intensity levels for a new driver with insufficient driving data by combining the Gaussian mixture model and the importance-weighted least-squares probabilistic classifier.
- [J5] [Li Zirui](#), et al. Interactive Behaviour Modelling for Vulnerable Road Users with Risk-Taking Styles in Urban

Scenarios: A Heterogeneous Graph Learning Approach. *IEEE Transactions on Intelligent Transportation Systems*, 2024, 25(8): 8538-8555. [Link] [IF=8.4]

Contribution: Early works overlooked the fact that different road users perceive the safety of the same interaction differently and also exhibit heterogeneous risk-taking styles. In this paper, we will develop a model for trajectory prediction based on risk-taking styles.

- [J6] **Li Zirui**, et al. ArrivalNet: Predicting City-wide Bus/Tram Arrival Time with Two-dimensional Temporal Variation Modelling. *IEEE Transactions on Intelligent Transportation Systems*, 2025. [Link] [IF=8.4]

Contribution: this paper proposes *ArrivalNet*, a two-dimensional temporal variation-based multi-step arrival time prediction for buses and trams. It decomposes the one-dimensional temporal sequence into intra-periodic and inter-periodic variations.

- [J7] **Li Zirui**, et al. Continual Driver Behaviour Learning for Connected Vehicles and Intelligent Transportation Systems: Framework, Survey and Challenges. *Green Energy and Intelligent Transportation*, 2023, 2(4): 100103. [Link] [34 citations] [IF=16.4]

Contribution: This paper presents a framework for continual driver behaviour learning (CDBL) by leveraging continual learning technology.

- [J8] **Li Zirui**, et al. Interactive Trajectory Primitives Representation and Extraction Based on Naturalistic Driving Data, *Automotive Engineering*, 46.8 (2024): 1382-1393. (In Chinese) [PDF]

Contribution: Taking interactive trajectory primitives as analysis units, a semantic-level representation and extraction method for multi-vehicle interactive behaviors is proposed.

- [J9] Gong H[#], **Li Zirui**[#], et al. Leveraging Multi-Stream Information Fusion for Trajectory Prediction in Low-Illumination Scenarios: A Multi-Channel Graph Convolutional Approach. *IEEE Transactions on Intelligent Transportation Systems*, 2023, 25(5): 3854-3869. [Link] [Code] [18 Github Stars] [IF=8.4]

Contribution: The paper proposes a novel approach for trajectory prediction in low-illumination scenarios by leveraging multi-stream information fusion, which integrates image, optical flow, and object trajectory information.

- [J10] Lin Y[#], **Li Zirui**[#], et al. Continual Interactive Behaviour Learning with Traffic Divergence Measurement: A Dynamic Gradient Scenario Memory Approach. *IEEE Transactions on Intelligent Transportation Systems*, 2023, 25(3): 2355-2372. [Link] [Code] [10 citations] [16 Github Stars] [IF=8.4]

Contribution: Most existing learning-based models trained with fixed datasets may perform poorly in continuously changing scenarios which is called “catastrophic forgetting”. To handle this problem, a novel continual learning approach for vehicle trajectory prediction is proposed.

- [J11] Xu, J[#], **Li Zirui**[#], et al. A Transferable Energy Management Strategy for Hybrid Electric Vehicles via Dueling Deep Deterministic Policy Gradient. *Green Energy and Intelligent Transportation*, 2022, 1(2): 100018. [Link] [Code] [37 citations] [92 Github Stars] [IF=16.4]

Contribution: This paper propose a novel transferable EMS by incorporating the DRL method and dueling network architecture for HEVs.

- [J12] **Li Zirui**, et al. Exploring the Over-stationarization in Deep Learning-based Bus/Tram Arrival Time Prediction: Analysis and Non-stationary Effect Recovery. *Under review at Information Fusion*. [Link] [IF=15.5]

Contribution: In this work, to trade off predictability and non-stationarity in the multi-step arrival time prediction of public transport, a new approach for multi-step ATP, named NSATP, is proposed. It consists of two parts: series stationarization and non-stationarity effect recovery. The first one aims at improving the predictability. As for the latter, a compensation module of over-stationarization is designed by learning scaling and shifting factors from raw data.

- [J13] **Li Zirui**, et al. Complementary Learning System Empowers Online Continual Learning of Vehicle Motion Forecasting in Smart Cities. *Under formal peer review at Nature Computational Science*. [Link][Code] [IF=18.3]

Contribution: This work introduces Dual-LS, a task-free, online continual learning paradigm for DNN-based motion forecasting, inspired by the complementary learning system of the human brain. It pairs two synergistic memory rehearsal replay mechanisms to accelerate experience retrieval while dynamically coordinating long-term and short-term knowledge representations.

- [J14] He, X[#], **Li Zirui**[#], et al. Reading minds on the road: Decoding perceived risk in automated vehicles through 140K ratings. *Under formal peer review at Proceedings of the National Academy of Sciences (PNAS)*. [Link] [Code and dataset] [Questionnaire] [Project Demo] [Result Demo] [IF=9.1]

Contribution: This study presents a novel method to decode people's perceived risk dynamics during

automated driving based on the largest perceived risk dataset to date, containing 141,628 discrete ratings and 236 hours of time-continuous data for realistic traffic.

- [J15] Lin Y[#], **Li Zirui**[#], et al. H2C: Hippocampal Circuit-inspired Continual Learning for Lifelong Trajectory Prediction in Autonomous Driving. *Under review at IEEE Transactions on Intelligent Transportation Systems*. [Link] [Code] [IF=8.4]

Contribution: As revealed by neuroscience, the hippocampal circuit plays a crucial role in memory replay, effectively reconstructing learned knowledge based on limited resources. Inspired by this, we propose a hippocampal circuit-inspired continual learning method (H2C) for trajectory prediction across varying scenarios.

- [J16] Yan, P[#], **Li Zirui**[#], et al. An Imputation-enhanced Hybrid Deep Learning Approach for Traffic Volume Prediction in Urban Networks: A Case Study in Dresden. *Data Science for Transportation*, 2024, 6(3): 22. [Link] [Platform]

Contribution: This paper proposes a hybrid deep learning approach for online traffic volume prediction in an urban network, which ensembles the long short-term memory (LSTM) neural network and the convolutional neural networks (CNN) in a parallel way.

- [J17] Li J, **Li Zirui**^{*}, et al. Experience-Shared Variable-Step Predictive Control of Range-Extended Electric Vehicles Using Transferable Driver Model. *IEEE Transactions on Intelligent Transportation Systems*, 2025, 26(1): 1123-1133. [Link] [IF=8.4]

Contribution: This paper leverages an experience-shared approach to variable-step MPC to improve range-extended electric vehicles energy efficiency, where a transferable driver model is designed to accommodate varying driver experience levels via knowledge transfer.

- [J18] Lin Y, ..., **Li Zirui**^{*}, et al. Escaping Stability-Plasticity Dilemma in Online Continual Learning for Motion Forecasting via Synergetic Memory Rehearsal. *Under review at IEEE Transactions on Neural Networks and Learning Systems*. [Link] [Code] [IF=8.9]

Contribution: this study proposes a novel CL method, synergetic memory rehearsal, for DNN-based motion forecasting. To ensure memory stability, it employs an inequality constraint that limits increments in the average loss over the memory buffer. Synergistically, a selective memory rehearsal mechanism is designed to enhance learning plasticity.

Key Conference Publications:

- [C1] **Li Zirui**, et al. An Ensemble Learning Framework for Vehicle Trajectory Prediction in Interactive Scenarios. *2022 IEEE Intelligent Vehicles Symposium (IV)*. IEEE, 2022: 51-57. [Link] [Code] [15 citations] [14 Github Stars]
- [C2] **Li Zirui**, et al. Transferable Driver Behaviour Learning via Distribution Adaption in the Lane Change Scenario. *2019 IEEE Intelligent Vehicles Symposium (IV)*. IEEE, 2019: 193-200. [Link] [31 citations]
- [C3] **Li Zirui**, et al. Development and Evaluation of Two Learning-based Personalized Driver Models for Pure Pursuit Path-tracking Behaviours. *2018 IEEE Intelligent Vehicles Symposium (IV)*. IEEE, 2018: 79-84. [Link] [33 citations]

Key Patents:

- [P1] Gong J, **Li Zirui**, et al. (2021) Interactive data reproduction method and system based on simulation platform. CN Patent ZL202111089628.X. (First authorship in student)[Link]
- [P2] Gong J, **Li Zirui**, et al. (2021) Coordinated control method and system for chassis movement and target strike of unmanned ground vehicles. CN Patent CN202111279251.4. (First authorship in student) [Link]

Other Co-author Publications and Preprints:

More than **20** Journal papers, with collaboration at top universities and institutes, including UC Berkeley, ETH Zurich, TU Delft, Tsinghua University, etc.

- [O1] Virtual-to-real knowledge transfer for driving behavior recognition: Framework and a case study. *IEEE Transactions on Vehicular Technology* 68.7 (2019): 6391-6402. [IF=7.1]
- [O2] Transfer learning for driver model adaptation in lane-changing scenarios using manifold alignment. *IEEE transactions on intelligent transportation systems* 21.8 (2019): 3281-3293. [IF=8.4]
- [O3] Hierarchical motion planning and tracking for autonomous vehicles using global heuristic based potential field and reinforcement learning based predictive control. *IEEE Transactions on Intelligent Transportation Systems* 24.8 (2023): 8304-8323. [IF=8.4]
- [O4] Reachability-based confidence-aware probabilistic collision detection in highway driving. *Engineering* 33

- (2024): 90-107. [IF=11.6]
- [O5] Beyond imitation: A life-long policy learning framework for path tracking control of autonomous driving. *IEEE Transactions on Vehicular Technology* 73.7 (2024): 9786-9799. [IF=7.1]
- [O6] Unravelling uncertainty in trajectory prediction using a non-parametric approach. *Transportation Research Part C: Emerging Technologies* 163 (2024): 104659. [IF=7.9]
- [O7] Efficient motion control for heterogeneous autonomous vehicle platoon using multilayer predictive control framework. *IEEE Internet of Things Journal* (2024). [IF=8.9]
- [O8] Ethical alignment decision-making for connected autonomous vehicle in traffic dilemmas via reinforcement learning from human feedback. *IEEE Internet of Things Journal* (2024). [IF=8.9]
- [O9] A transferable perception-guided EMS for series hybrid electric unmanned tracked vehicles. *Energy* 306 (2024): 132367. [IF=9.4]
- [O10] Cross-Scenario End-to-End Motion Planning in Off-Road Environment: A Lifelong Learning Perspective. *IEEE Robotics and Automation Letters* (2025). [IF=5.3]
- [O11] Hierarchical Path Planning and Motion Control Framework Using Adaptive Scale Based Bidirectional Search and Heuristic Learning Based Predictive Control. *IEEE Transactions on Vehicular Technology* (2025). [IF=7.1]
- [O12] Enhancing the Collaborative Decision-Making Performance of Connected and Autonomous Vehicles: A Multi-Modal Failure-Aware Graph Representation Approach. *IEEE Transactions on Intelligent Transportation Systems* (2025). [IF=8.4]
- [O13] A naturalistic trajectory dataset with dense interaction for autonomous driving. *Scientific Data* 12.1 (2025): 1084. [IF=8.7]
- [O14] Effective contact texture region aware pavement skid resistance prediction via convolutional neural network." *Computer-Aided Civil and Infrastructure Engineering* 39.13 (2024): 2054-2070. [IF=9.1]
- [O15] Personalized lane change planning and control by imitation learning from drivers. *IEEE Transactions on Industrial Electronics* 70.4 (2022): 3995-4006. [IF=7.2]
- [O16] How far ahead should autonomous vehicles start resolving predicted conflicts? Exploring uncertainty-based safety-efficiency trade-off. *IEEE Transactions on Intelligent Transportation Systems* 25.10 (2024): 14183-14195. [IF=8.4]
- [O17] MV-STGHAT: Multi-View Spatial-Temporal Graph Hybrid Attention Network for Decision-Making of Connected and Autonomous Vehicles. *IEEE Transactions on Vehicular Technology* (2024). [IF=7.1]
- [O18] Occlusion handling and multi-scale pedestrian detection based on deep learning: A review. *IEEE Access* 10 (2022): 19937-19957. [IF=3.6]
- [O19] Learning-based hierarchical decision-making framework for automatic driving in incompletely connected traffic scenarios. *Sensors* 24.8 (2024): 2592. [IF=3.5]
- [O20] Fusion of gaze and scene information for driving behaviour recognition: A graph-neural-network-based framework. *IEEE Transactions on Intelligent Transportation Systems* 24.8 (2023): 8109-8120. [IF=8.4]
- [O21] Graph reinforcement learning-based decision-making technology for connected and autonomous vehicles: Framework, review, and future trends. *Sensors* 23.19 (2023): 8229. [IF=3.5]
- [O22] Spatio-Temporal Coordinated Decision-Making in Heterogeneous Platooning via Interflow Hub-and-Spoke Graph Reinforcement Learning. *IEEE Internet of Things Journal* (2025). [IF=8.9]
- [O23] Cooperative Autonomous Driving in Diverse Behavioral Traffic: A Heterogeneous Graph Reinforcement Learning Approach. *arXiv preprint arXiv:2509.25751* (2025).
- [O24] Multilevel Graph Reinforcement Learning for Consistent Cognitive Decision-making in Heterogeneous Mixed Autonomy. *arXiv preprint arXiv:2408.08516* (2024).
- [O25] Decision-making technology for autonomous vehicles: Learning-based methods, applications and future outlook. *2021 IEEE International Intelligent Transportation Systems Conference (ITSC)*. IEEE, 2021.
- [O26] Autonomous driving strategies at intersections: Scenarios, state-of-the-art, and future outlooks. *2021 IEEE International Intelligent Transportation Systems Conference (ITSC)*. IEEE, 2021.

Selected Projects

During my doctoral studies, I participated in multiple collaborations and I developed models and algorithms validated across various commercial terminals and platforms.

Project 1: Interactive Motion Planning for Automated Vehicles (Ph.D. Project)

Description: This work focuses on modelling the interactive behaviours among multiple traffic participants, quantifying and integrating various types of risk, and designing motion planning algorithms. Accordingly, it

proposes a spatiotemporal risk continual learning–based interactive motion planning method for unmanned vehicles. The “behaviour prediction–risk assessment–motion planning” integrated approach aims to enhance driving safety in dynamically interactive environments through accurate spatiotemporal risk assessment.

Outcome: [\[Project website\]](#) [\[Paper J1\]](#) [\[Paper J3\]](#) [\[Paper J5\]](#) [\[Book 1\]](#) [\[Book 2\]](#) [\[Real Vehicle Demo\]](#)

Project 2: Large-scale Driver Perceived Risk analysis with 140K+ ratings

Description: Driver's perceived risk in self-driving cars can create the very danger that automation is meant to prevent: a frightened rider may hesitate when seconds matter, misjudge hazards, or disengage. However, how this risk evolves dynamically in real world remains poorly understood. Understanding what sparks that fear is therefore critical to real-world safety. We conducted the largest controlled investigation of passenger emotion in automated driving to date. 2,164 volunteers viewed short, high-fidelity videos of four common highway situations and provided 141,628 second-by-second safety ratings. **This project is partly supported by EU H2020 funding. Complete data collection knowledge and data management experience.**

Outcome: [\[Paper J14\]](#) [\[Code and dataset\]](#) [\[Questionnaire\]](#) [\[Project Demo\]](#) [\[Result Demo\]](#)

Project 3: Public Transport Arrival Time Prediction

Description: Accurate arrival time prediction (ATP) of buses and trams plays a crucial role in public transport operations. Current methods only focused on modelling the time series along the temporal dimension by investigating the point-to-point relationship between different timesteps. This strategy overlooked the latent periodic information within time series. Moreover, most studies developed algorithms for ATP based on a single or a few routes of public transport, which reduces their applicability in public transport management systems. To this end, this work proposes ArrivalNet, a two-dimensional temporal variation-based multi-step ATP for buses and trams. 125 days of public transport data from Dresden were collected for model training and validation. This is a cooperation with [DVB](#).

Outcomes: [\[Paper J6\]](#) [\[Paper J12\]](#) [\[Paper J16\]](#) [\[Traffic Monitor Platform\]](#)

Project 4: A naturalistic trajectory dataset with dense interaction

Description: This work presents InterHub, a curated dataset of dense multi-agent interaction events, derived from large-scale naturalistic driving recordings. The dataset is accompanied by an open-source toolkit that enables users to expand InterHub by mining additional interaction events from both public and private driving data. By offering a unified taxonomy, rich annotations, and extensible tools, InterHub supports diverse research needs—from interaction behaviour modelling to algorithm benchmarking—and promotes reproducibility, scalability, and cross-dataset comparison in autonomous driving studies.

Outcome: [\[Nature Scientific Data\]](#) [\[Video\]](#) [\[Code\]](#)

Project 5: Autonomous driving decision-making technology based on road-side enhancement

Description: The project advances cross-domain reinforcement techniques for autonomous driving clusters, aiming to address the fundamental challenge of developing operational mechanisms and collaborative risk modelling and control theories. It seeks to design and implement an integrated autonomous driving cluster system that unifies perception, communication, computation, and decision-making. To meet the demands of complex traffic environments and enhance traffic efficiency, the project adopts a three-tier design framework—system integration, modular synergy, and holistic optimization. Within this framework, it will investigate key technologies for cluster-level autonomous driving and establish mechanisms for cross-domain reinforcement in cluster operations. Research efforts will concentrate on group-level performance modelling and optimization of vehicle clusters, near- and long-range collaborative modelling and decision-making for traffic clusters, cross-domain environmental perception, and reinforcement learning–driven behavioural modelling and co-evolution of clusters. The project aims to realize a robust design of cross-domain reinforced autonomous driving clusters. The target system is expected to be scalable for future large-scale deployment in real-world traffic environments, with breakthroughs in safety, reliability, stability, and extensibility. This is an Industrial Cooperation with [Baidu Apollo](#).

Outcome: [\[News\]](#)

Project 6: Smart Mobility & Delivery

Description: As a Postdoctoral Researcher at Nanyang Technological University (NTU) within the Future-Oriented Continental's Urban Society (FOCUS) Lab, I work on advancing safe, reliable, and efficient robotic and navigation technologies for delivery robots, autonomous vehicles, and electric vehicles. My research focuses on enabling robust operation in complex urban environments and diverse weather conditions through the development of situation awareness, interaction-aware behaviour modelling, and decision-making algorithms. This involves multimodal sensor fusion, perception, and trajectory prediction, combined with test rig set-up and experimental validation. The project contributes to building scalable autonomous mobility solutions that improve safety, efficiency, and sustainability in future urban transportation systems. This is the postdoc project at NTU, and

cooperation with [Continental](#).

Project 7: Lifelong Learning Research Project

Description: AI underpins smart city services, yet deep neural networks (DNN) that forecast vehicle motion struggle with catastrophic forgetting, the loss of earlier knowledge when models are updated. Conventional fixes enlarge the training set or replay past data, but these strategies incur high data collection costs, sample inefficiently and fail to balance long- and short-term experience, leaving them short of human-like continual learning. We introduce Dual-LS, a task-free, online continual learning paradigm for DNN-based motion forecasting that is inspired by the complementary learning system of the human brain. Dual-LS pairs two synergistic memory rehearsal replay mechanisms to accelerate experience retrieval while dynamically coordinating long-term and short-term knowledge representations. Tests on naturalistic data spanning three countries, over 772k vehicles and cumulative testing mileage of 11,187 km show that Dual-LS mitigates catastrophic forgetting by up to 74% and reduces computational resources by 94%. Meanwhile, it endows DNN-based vehicle motion forecasting with computation efficient and human-like continual learning adaptability fit for smart cities. **A paper is under review at [Nature Computational Science](#) as the extension of the doctoral thesis.**

Outcomes: [[Paper J10](#)] [[Paper J13](#)] [[Paper J5](#)] [[Paper J18](#)]

Awards

My research has been consistently recognized through national and international awards, including highly cited publications, prestigious fellowships, and innovation prizes. These achievements highlight the impact and originality of my work in intelligent vehicles, as well as strong project coordination and leadership skills.

- [**Highlight**] NTU AI-for-X research grant 250,000 SGD
- [**Highlight**] 2025 Marie Skłodowska-Curie Postdoctoral Fellowship (Success rate 8.4%, hosted by EPFL)
- [**Highlight**] NTU AI-for-X Postdoctoral Fellowship (Success rate < 1%)
- [**Highlight**] SAE China Best Doctoral Dissertation Award (Highest honour for Ph.D. in vehicle engineer)
- Excellent Talk at China Society of Automotive Engineers Doctoral Student Forum 2025 [[News](#)]
- BIT Excellent Doctoral Dissertation Seedling Fund (10,000 CNY) [[News](#)]
- China Association for Science and Technology Youth Talent Support (40,000 CNY)
- The 3rd Place Prize in ROAD mode of TRA VISIONS 2024 Young Researcher Competition [[News](#)]
- 2024 IEEE ITSS New Initiatives with proposal “The Development of Guide for Driving Risk Assessment Procedure in SAE Level 1 to 3” (6,000 USD)
- 2024 IEEE ITSS New Initiatives with proposal “Evidence-Based Communication Channel for Autonomous Driving Technology in the IEEE ITS Community” (8,000 USD)
- Fellowship for Promoting Women, Young Professionals and Leadership in ITS (2023 and 2025)
- Two-Year Graduate Student Fellowship of China Scholarship Councils (CSC)
- Excellent Graduate Study Scholarship (First class, twice, 2021 and 2024)
- National Scholarship for College Student, Ministry of Education of China (Twice, 2021 and 2024)
- NORINCO Scholarship
- Champions for Formula Student Driverless China, 2017&2018 [[Video 1](#)][[Video 2](#)][[Video 3](#)][[News](#)]
- First Prize Innovation Scholarship, Ministry of Industry and Information Technology 2018
- Highly Cited and high PSCI Paper Recognized by CNKI for “Review on Machine Learning Methods for Motion Planning and Control Policy of Intelligent Vehicles (In Chinese)”.

Professional Activities

Journal Reviewer

- IEEE Transactions on Intelligent Transportation Systems, IF=8.4, JCR Q1.
- IEEE Transactions on Neural Networks and Learning Systems, IF=8.9, JCR Q1.
- IEEE Transactions on Pattern Analysis and Machine Intelligence, IF=18.6, JCR Q1.
- IEEE Transactions on Intelligent Vehicles, IF=14, JCR Q1.
- IEEE Transactions on Vehicular Technology, IF=7.1, JCR Q1.
- IEEE Transactions on Transportation Electrification, IF=8.3, JCR Q1.
- IEEE Internet of Things Journal, IF=8.9, JCR Q1.
- Accident Analysis & Prevention, IF=6.2, JCR Q1.
- Transportation Research Part C: Emerging Technologies, IF=7.9, JCR Q1.

- Transportation Research Part F: Traffic Psychology and Behaviour, IF=4.4, JCR Q1.
- IEEE Robotics and Automation Letters, IF=5.3, JCR Q1.
- IET Intelligent Transport Systems, IF=2.5, JCR Q1.
- Frontiers of Information Technology & Electronic Engineering, IF=2.9, JCR Q.
- IEEE/CAA Journal of Automatica Sinica, IF=19.2, JCR Q1.

Conference Reviewer

- IEEE Intelligent Vehicles Symposium, Flagship conference in ITSS
- IEEE Conference on Intelligent Transportation Systems, Flagship conference in ITSS
- Transportation Research Board (TRB) Annual Meeting
- IEEE/RSJ Conference on Intelligent Robots and Systems, Flagship conference in RAS
- IEEE International Conference on Robotics & Automation, Flagship conference in RAS

Editorial Activities

- Automotive Engineering, Editor Assistant
- IET Intelligent Transport Systems, Guest Editor
- Actuators, Guest Editor
- Frontiers in Robotics and AI, Guest Editor

Invited Talks

- [Mar. 2026] Towards Interactive, Trustworthy and Self-evolving Autonomy: From Embedded AI to Embodied AI in Prof. Wen-Hua Chen's group at PolyU, HK.
- [Oct. 2025] Research on Interactive Motion Planning Method of Automated Vehicles based on Spatiotemporal Risk Continual Learning at the lecture talk in NTU.
- [Oct. 2025] Research on Interactive Motion Planning Method of Automated Vehicles based on Spatiotemporal Risk Continual Learning at AutoMan Lab in NTU.
- [Jul. 2025] Research on Interactive Motion Planning Method of Automated Vehicles based on Spatiotemporal Risk Continual Learning at Prof. [Yuanqing Xia](#)'s Group in BIT.
- [Jul. 2025] Research on Interactive Motion Planning Method of Automated Vehicles based on Spatiotemporal Risk Continual Learning at China Society of Automotive Engineers Doctoral Student Forum.
- [Apr. 2024] ArrivalNet: Multi-step Bus/tram Arrival time Prediction at 2024 TUD-KTH joint workshop.
- [Nov. 2023] Towards integrating subjective and objective risks in human-centric automated driving at 2023 ITS WiE/YP Workshop & Research Forum, Cairo, Egypt.
- [Sep. 2023] Continual Driver Behaviour Learning in Urban Scenarios at ITSC 2023, Bilbao, Bizkaia, Spain.
- [Dec. 2022] Interactive Behaviour Modelling for Vulnerable Road Users with Risk-taking Styles in Urban Scenarios: A Heterogeneous Graph Learning Approach at IEEE ITSC 2022 workshop.
- [Dec. 2022] Towards Safe, efficient and Co-operative Decision-making for CAVs in Mixed Autonomy: An Attention-Enhanced Graphic Reinforcement Learning Approach at MFTS 2022.
- [Nov. 2022] Interactive Behaviour Modelling for Vulnerable Road Users with Risk-taking Styles: A Heterogeneous Graph Learning Approach at IEEE ITSC 2022 workshop Cooperative and Automated Driving.
- [Oct. 2022] Graph Convolution-Based Deep Reinforcement Learning for Multi-Agent Decision-Making in Mixed Traffic environments at IEEE ITSC 2022.
- [Jun. 2022] Interactive Behaviour Modelling for Vulnerable Road Users with Risk-taking Styles: A Heterogeneous Graph Learning Approach at the Chair of Traffic Process Automation TU Dresden.
- [Oct. 2021] Defining the Interactive Behaviour: A Mutual Information-based Approach at hEAT Lab, TU Delft.
- [Jun. 2021] Graph Neural Network for Interactive behaviours: Modelling, Risk Assessment and Multi-task Prediction at Traffic Dynamics, Modelling and Control lab, TU Delft.
- [Apr. 2021] Graph Neural Network for Interactive behaviours: Modelling, Risk Assessment and Multi-task Prediction at hEAT Lab, TU Delft.
- [Jun. 2019] Transferable Driver Behaviour Learning via Distribution Adaption at IEEE IV 2019.
- [Jun. 2018] Learning-Based Personalized Driver Model for Path-Tracking Behaviours at IEEE IV 2018.

(Co-)Chair

I chaired and co-chaired sessions at leading conferences such as IEEE ITSC and IV, coordinated invited speakers, and served on organizing committees, including MFTS 2022. These experiences demonstrate my ability to lead dissemination activities, manage international collaborations, and amplify the impact of research communities.

- Associate Editor for CIS-RAM 2026
- Associate Editor for IEEE ITSC 2026
- Co-chair, ITSC 2026 Workshop “AI-Empowered Human-Machine Interaction for Intelligent Vehicles”
- Chair, Foundation Model-Empowered Decision-making and Planning for Intelligent Mobility Systems, at IEEE ITSC 2026
- Co-chair, Next-Generation Autonomous Driving Systems: Self-Evolving Decision-Making, End-to-End Architectures, and Data-Driven Evaluation, at IEEE ITSC 2026
- Organizer, CSAE Series Talk
- Working group leader, intelligent vehicle risk assessment under IEEE Standard Association
- Co-chair, Cooperative Decision-making for Connected and Automated Vehicles in ITS at IEEE ITSC 2023
- Associate Editor for IEEE IV 2023
- Chair, Social, interactive and safe behaviours for AVs: benchmarks, models and applications at IEEE IV 2023
- Organising committee and Proceedings editor, [MFTS 2022](#)
- Chair, Social and Interactive Behaviour Modelling in ITS at IEEE ITSC 2021

Mentorship

Beijing Institute of Technology:

[Cheng Gong](#), [Yunlong Lin](#), Chenxu Wen, [Haowen Wang](#), [Hailong Gong](#), Lianzhen Wei, [Yangtian Yi](#), [Xianqi He](#), [Jingyi Xu](#), Yingqi Tan, [Tanran Chen](#).

TU Dresden:

Fanyi Wei with Diploma thesis: Optimization of Velocity for Eco-Driving using Reinforcement Learning based on Simulation Data of BMW i3 (Co-with Prof. [Meng Wang](#))

External mentor (QMUL, Co-with Prof. [Xinwei Wang](#)):

Ziyue Tang (MSc)

Yixuan Li (Ph.D. student): Intelligent Routing and Scheduling for Airport Airside Operations [Loughborough Uni.]

Yueru Shang (Ph.D. student): Intelligent Transport Network Optimisation [Loughborough Uni]

Jiaxing Li (Ph.D. student): Towards Intelligent Decision Support and Risk Assessment Systems [QUMIL]

Lecturer Experience

TU Dresden: Python course at TU Dresden: Slides preparation, lecturer, exercise.

Nanyang Technological University: Lecturer talk at NTU in Prof. Lyu’s course.